Sentinel · Fraud Detection Wallet

Fraud Patterns in Plain English

What makes the system call a transaction suspicious, why it stops some payments, and a step-by-step kitchen of tests you can run yourself — no code required.

DOCUMENT patterns.pdf	VERSION 1.0
DATE 30 October 2026	AUDIENCE Owners, testers, reviewers
PLATFORM Web app, USSD and admin console	READING TIME About 12 minutes

What is inside

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The one-minute version. Every transfer and withdrawal passes five quick behaviour checks, then a set of hard rules, then a machine-learning score. Behaviour checks and rules can stop a payment outright; the ML score mostly decides how suspicious it looks afterwards. Money that moves also earns the business a fee — 0.8% on transfers, 1.1% on withdrawals — and that fee is the revenue shown on the admin dashboard.

SECTION 1

How one payment is screened

Think of the system as a security guard standing between the customer and the money. The guard asks the same questions in the same order every time, so the same suspicious behaviour always produces the same answer.

1 · BEHAVIOUR
Have you been unusually busy in the last few minutes?

2 · PLACE
Are you suddenly far away from where you last paid?

3 · BASICS
Is the amount valid, is the receiver real, is the money there?

4 · BRAIN
The ML model scores the payment from 0 to 100.

5 · RULES
Hard rules can override the score and stop the payment.

What happens at each step

Step	What it looks at	Can it stop the payment?
Behaviour checks	Counts and totals over 5–10 minute windows: how many large payments, how many different people, how many failed withdrawals.	Yes — scores 85 or 90
Location	Distance between the last city used and the current one, within 5 minutes. Adds points to the risk score.	Not alone; feeds the rules
Validation	Amount above zero, receiver exists and is approved, balance covers amount *plus fee*.	Rejects the request
ML score	Eleven features such as amount, time of day, account age and recent activity.	No — informs review
Rules	Known danger combinations: odd hour, huge amount, almost-empty account, recent failed attempts.	Yes — scores 75 to 90

Blocked attempts still count. If a payment is stopped, the attempt is stored in the history like any other transaction. That means trying again immediately keeps you inside the same 5- or 7-minute window — useful to know when testing, and deliberately strict for a real attacker.

What the customer sees

Blocked

A red panel replaces the form, with the reason, the risk score and the level, for example “Risk Score: 85% · Risk Level: HIGH”, plus a “contact support” line. The customer also gets an SMS saying the money did not move.

Passed, but suspicious

The payment completes and may show a location warning. If the score lands between 41 and 70 the customer is told the transaction was flagged for review, and the admin sees it in the fraud alerts.

SECTION 2

Risk scores, levels and alerts

A score is a percentage: 0 is “nothing odd here”, 100 is “stop everything”. The level is just the score in words, and each surface of the product uses a different cut-off.

Band	Level	What happens	Where you see it
0 – 30	LOW	Payment proceeds normally.	Risk column in transaction lists.
31 – 70	MEDIUM	Proceeds, unless a rule blocks it. A successful payment scoring 41–70 is “flagged for review”.	Receipt note, admin alerts if the score is above 50.
71 – 100	HIGH	Any rule that fires also blocks. Pattern blocks always land here.	Fraud alerts, high-risk counter.

The two admin queues

Queue	Rule	Where
Fraud alerts	Prediction is FRAUD *and* score is above 50, reviewed or not.	Admin → Fraud Alerts
AI review queue	Prediction is still PENDING with a score above 0.	Dashboard KPI and counters

> **Blocked is not the same as fraud.** A blocked row means “we refused to move the money”. A row can also complete with a FRAUD prediction when no blocking rule fired — that is exactly the case the review queue exists for.

Money rules that apply to every payment

Rule	Value	Effect
Transfer fee	0.8%	Added to the sender's debit; the receiver gets the full amount.
Withdrawal fee	1.1%	Added to the amount the customer pays to take cash out.
Minimum withdrawal	TSh 1,000	Smaller withdrawals are rejected before scoring.
Balance needed	amount + fee	Otherwise the transfer is refused; a failed withdrawal counts as an attempt.
Receiver must be approved	—	Pending or inactive accounts cannot receive money.
No self-transfer	—	A customer cannot pay their own card.

Fees are stored on each transaction row and only count as revenue when the row is SUCCESS. The admin dashboard shows “Revenue Today” using the Dar es Salaam calendar day.

SECTION 3 · BEHAVIOUR PATTERNS

Patterns 1 to 3: too much, too fast

These three checks run before anything else and stop the payment outright. They look only at the last few minutes and at how many distinct people are involved, never at who the customer is.

#	Nickname	What the system counts	Blocked when	Score
1	**The big spender** Large volume	Payments larger than TSh 1,000,000 in the last 5 minutes: how many, and their total.	3 or more *and* total above TSh 10,000,000	85
2	**Spray and pray** Many receivers	Different people you have paid in the last 5 minutes.	The 4th different receiver	90
3	**The money mule** Many senders	Different people who have paid the *receiver* in the last 7 minutes.	The 3rd different sender	90

Why these numbers

- **Pattern 1** catches someone draining an account in a burst: two payments of 4,000,000 are fine, but a third pushes the window past 10,000,000 and the count to 3.
- **Pattern 2** catches a stolen card being used to cash out to many hands quickly. Paying the same person again does not count twice — only distinct receivers do.
- **Pattern 3** catches a collection account: three different people pushing money into one wallet in seven minutes looks like a mule, so the third *sender* is stopped.

> **These blocks ignore the amount band.** Pattern 2 fires on the fourth receiver even for TSh 1,000 payments, and Pattern 3 on the third sender. That is intentional: the shape of the behaviour matters more than the size.

SECTION 4

Where you are, and how often you fail

The teleport problem

The web app never asks for a city. On submit it reads the browser's coordinates silently, finds the nearest supported city, and compares it with the city of your last located payment in the previous 5 minutes. If the two differ, points are added to the risk score:

Change	Points	Example
Any different city	+10	Dar es Salaam → Morogoro
More than 50 km	+15	Dar es Salaam → Morogoro (≈180 km)
More than 200 km	+30	Dar es Salaam → Dodoma
More than 500 km	+50	Dar es Salaam → Mwanza (≈843 km)

A change alone does not block anything — but with a payment above TSh 500,000 it triggers the rule “suspicious location change”, and the customer sees a warning such as “Location changed from Dar es Salaam to Mwanza (843 km away) in 5 minutes”.

> **Graceful fallbacks.** If the customer denies the location prompt, the system keeps their last known city. If the coordinates are more than 250 km from every supported city, the reading is treated as “not in a scored city” and the last known city is kept too. A customer is never blocked because a GPS fix failed.
>
> **USSD** has no GPS, so it always uses the customer's last known city. Location patterns cannot be triggered over USSD — use the web app.

The locked wallet

Trying to withdraw more than you have is recorded as a failed attempt. Three failed attempts lock withdrawals and also make transfers look dangerous:

Situation	Result	Score
1st failed withdrawal	Refused; attempt 1 of 3 recorded.	60
2nd failed withdrawal	Refused; attempt 2 of 3 recorded.	70
3rd failed withdrawal	Refused; withdrawals now blocked.	80
Any withdrawal afterwards	Blocked with “temporarily blocked due to multiple failed attempts”.	95
Any transfer afterwards	Blocked while the attempt counter is at 3 or more (10-minute window).	85

How the lock lifts: one successful withdrawal resets the counter, or 24 hours pass without a new failed attempt. A blocked attempt does not extend the 24-hour clock by itself; the counter is reset lazily when the next withdrawal is checked.

SECTION 5

Hard rules and the machine-learning score

Rules that always block

These override the ML result. If any rule matches, the payment becomes FRAUD / HIGH and is stopped; when several match, the highest score wins and the last reason is shown.

Transfers

Rule	Condition	Score
Recent failed withdrawals	3 or more failed withdrawals in the last 10 minutes	85
Odd-hour large transfer	Before 06:00 East Africa Time and amount above 1,000,000	80
Location jump with a large amount	Location score above 30 and amount above 500,000	75
Extreme amount	Amount above 10,000,000	90

Withdrawals

Rule	Condition	Score
Burst of activity	3 or more transactions in 5 minutes totalling more than 10,000,000	85
Odd-hour large withdrawal	Before 06:00 East Africa Time and amount above 1,000,000	85
Location jump with a large amount	Location score above 30 and amount above 500,000	80
Emptying the account	Withdrawing more than 90% of the balance when above 1,000,000	80
Extreme amount	Amount above 10,000,000	90

What the ML model actually does

The Python service loads a trained XGBoost model. The API hands it eleven numbers describing the moment: the amount, how many transactions and how much money in the last 5 minutes, unique receivers and senders, failed withdrawals in the last 10 minutes, the hour of day, the customer's average amount, how far this amount deviates from that average, the account's age in days, and the balance. The model returns a probability, which becomes the 0–100 score.

If the model is down

The API falls back to score 0, LEGIT, LOW and keeps working. Rules and behaviour patterns still block. The health endpoint reports whether the ML service is reachable.

How to spot a real ML decision

Look at the risk score in the admin transaction row or the customer receipt. Pattern and rule blocks are round numbers (75–100) with a reason; an ML decision tends to be an odd number like 37 or 62 without a rule reason.

SECTION 6

Test kitchen

Before you start. You need one admin and two to five customer accounts. The admin can create each customer, set an opening balance and approve them (Admin → Users → Create User), or approve accounts that registered themselves. Cards are issued automatically: the sender types the receiver's 16-digit number, which the receiver can read on their dashboard.

> **Two practical tips.** The windows are real time, so keep a clock handy — five minutes pass faster than you think. And a blocked attempt is recorded, so if a test blocks earlier than expected, wait out the window before the next one instead of piling on attempts.

Recipe 1 — The big spender (Pattern 1)

expected: blocked, 85

Set up: one sender with at least TSh 11,100,000 (amounts plus fees) and three approved receivers.

1. Send **4,000,000** to receiver A. Expect success; fee 32,000.
2. Within five minutes, send **4,000,000** to receiver B. Expect success; fee 32,000.
3. Still inside five minutes, send **3,000,000** to receiver C.

Result: the third payment is blocked with score 85 and "Large transaction volume — 3 large transactions totaling TSh 11,000,000.00 in 5 minutes". The receiver is not credited and the sender's balance is unchanged by the blocked attempt.

Recipe 2 — Spray and pray (Pattern 2)

expected: blocked, 90

Set up: one sender and four approved receivers, small amounts.

1. Send **10,000** to receiver A, then **10,000** to B, then **10,000** to C — all within five minutes.
2. Send **10,000** to a fourth receiver, D.

Result: the fourth payment is blocked with score 90 and "You have sent money to 3 different recipients in the last 5 minutes. Maximum allowed: 3 recipients per 5 minutes." Paying A again would not have triggered it.

Recipe 3 — The money mule (Pattern 3)

expected: blocked, 90

Set up: one receiver, R, and three senders A, B, C, all with balance.

1. A sends **20,000** to R.
2. B sends **20,000** to R, within seven minutes.
3. C sends **20,000** to R, still within seven minutes.

Result: C's payment is blocked with score 90 and "The recipient has received money from 2 different senders in the last 7 minutes." The third *sender* is the one stopped, not the receiver.

Recipe 4 — The teleport (location + rule)

expected: blocked, 75

Set up: a sender with at least TSh 1,300,000. Use Chrome DevTools to fake the location: **F12** → ⋮ **menu** → **More tools** → **Sensors** → **Location** → **Custom location**. Approve the browser's location prompt the first time.

1. Set the location to Dar es Salaam: latitude `-6.7924`, longitude `39.2083`. Send **600,000** — expect success.
2. Within five minutes change the location to Mwanza: `-2.5164`, `32.9171`. Send **600,000** to another receiver.

Result: the second payment is blocked with score 75 and "Suspicious location change with large transaction". The location component is 60 points (10 for any change + 50 for more than 500 km). A smaller second payment, say TSh 50,000, would succeed but show the warning "Location changed from Dar es Salaam to Mwanza (843 km away) in 5 minutes" on the result and the receipt.

Recipe 5 — The locked wallet

expected: withdrawal 95, then transfer 85

Set up: a customer with TSh 1,000,000.

1. Try to withdraw **2,000,000** three times (the amount exceeds the balance).
2. Try to withdraw **50,000**.
3. Try to send **10,000** to anyone.

Result: the three attempts report failed attempts 1, 2 and 3 and the third says withdrawals are blocked. The 50,000 withdrawal is blocked with 95 and "Your withdrawals are temporarily blocked due to multiple consecutive failed withdrawal attempts". The transfer is blocked with 85 and "Multiple failed withdrawal attempts detected". Waiting 24 hours, or making one successful withdrawal, clears it.

Recipe 6 — Odd hours, huge amounts, empty accounts

expected: 80 / 85 / 90 / 80

Set up: a customer with a large balance. Change the system clock only if you are testing locally; in production simply try before 06:00 East Africa Time.

1. Before 06:00, send **1,500,000** → blocked, 80 ("Abnormal transaction at unusual hour with large amount").
2. Before 06:00, withdraw **1,500,000** → blocked, 85 (withdrawal version of the same rule).
3. Send **10,000,001** → blocked, 90 ("Transaction amount exceeds security threshold").
4. Set the balance to 1,600,000 and withdraw **1,500,000** → blocked, 80 ("Attempting to withdraw more than 90% of balance").

Note: each of these needs the balance to cover amount plus fee, or the request is rejected for insufficient funds before the rules are reached.

Recipe 7 — Where the score shows up

no block required

1. Make a few ordinary payments of different sizes at a normal hour.
2. Open **Admin** → **Fraud Alerts**. Any FRAUD prediction scoring above 50 appears here, newest first.
3. Open a user's **Dashboard**: the fraud card reports analysed / fraud / average / highest, and a successful payment scoring 41–70 is marked “flagged for review” on the receipt.
4. In Fraud Alerts, press **Mark Reviewed** — the row leaves the unreviewed filter and the alert counter drops.

Recipe 8 — Fees and revenue

expected: 800 and 550

1. Send **100,000**: the form shows fee 800.00 and total 100,800.00; the receiver gets 100,000.
2. Withdraw **50,000**: fee 550.00, total debited 50,550.00.
3. Open the admin dashboard: **Revenue Today** increases by 1,350.00.

Note: blocked rows store their fee but never count as revenue; only SUCCESS rows do.

SECTION 7

Automated tests and troubleshooting

Run the whole fraud suite

`pnpm --filter @fraud/api test` runs every API test. To run one file: `pnpm --filter @fraud/api exec vitest run test/integration/transfer.test.ts`

File	What it proves
`test/unit/patterns.test.ts`	Patterns 1–3 fire at exactly 3 large / 10M, the 4th receiver and the 3rd sender, and not before.
`test/unit/rules.test.ts`	Every rule's condition, score and reason, including rule-overrides-ML.
`test/unit/location.test.ts`	City distances and the 10 / 15 / 30 / 50 point bands.
`test/unit/features.test.ts`	The eleven ML features are computed as the model expects.
`test/unit/mlClient.test.ts`	Timeouts and errors fall back to LEGIT instead of failing the payment.
`test/integration/transfer.test.ts`	End-to-end transfers, pattern blocks, rule blocks, fees, atomic debits.
`test/integration/withdrawal.test.ts`	Withdrawal rules, failed-attempt counter and lockout.
`test/integration/velocity.test.ts`	Window counters, aborted rows counting, 24-hour reset, overdraft guard.
`test/integration/ussd.test.ts`	The same rules and ledger through the phone channel.
`test/integration/adminMonitoring.test.ts`	Alerts, counters, review queue and daily revenue.

When a test does not behave

Symptom	Most likely cause
Nothing is blocked	The payment failed validation first: receiver not approved or inactive, amount above balance plus fee, or sending to your own card.
Pattern 2 fired while testing Pattern 1	You used four different receivers. Use three, or repeat a receiver.
The location rule never fires	Location permission denied, both payments from the same city, more than five minutes apart, or the amount is 500,000 or less.
Location rule fires when it should not	The reading was more than 250 km from every supported city, so the previous city was kept — check the test coordinates against the city list.
Withdrawals still locked	The counter resets after 24 hours or on one successful withdrawal; blocked attempts do not reset it.
No alert in the admin list	The row scored 50 or less, or you are filtering by unreviewed/reviewed status.
Scores are always 0 and LEGIT	The ML service is not running. Rules still work; start the model service and check the API health output.

Symptom	Most likely cause
Fee seems to make the payment fail	The balance must cover amount *plus* 0.8% (transfer) or 1.1% (withdrawal).

> **Resetting between experiments.** Every window is time based, so waiting is the cleanest reset: five minutes clears patterns 1 and 2 and the location comparison, seven minutes clears pattern 3, ten minutes clears the failed-withdrawal rule. For a clean database, use a fresh test database rather than deleting rows from the live one.

SECTION 8

Cheat sheet

#	Trigger	Threshold	Score
1	Large volume	3+ payments over 1,000,000 totalling more than 10,000,000 in 5 minutes	85
2	Many receivers	4th distinct receiver in 5 minutes	90
3	Many senders	3rd distinct sender to one receiver in 7 minutes	90
4	Odd-hour transfer	Before 06:00 EAT and above 1,000,000	80
5	Odd-hour withdrawal	Before 06:00 EAT and above 1,000,000	85
6	Location jump, transfer	Location score above 30 and above 500,000	75
7	Location jump, withdrawal	Location score above 30 and above 500,000	80
8	Extreme amount	Above 10,000,000 (both products)	90
9	Withdrawal burst	3+ transactions and over 10,000,000 in 5 minutes	85
10	Almost empty account	More than 90% of balance and above 1,000,000	80
11	Failed withdrawals	3 failed attempts, then transfers blocked for 10 minutes	85
12	Withdrawal lockout	3 failed attempts; lifts after 24 hours or one success	95

Numbers worth remembering

Money

- Transfer fee **0.8%**, paid by the sender.
- Withdrawal fee **1.1%**, paid by the customer.
- Minimum withdrawal **TSh 1,000**.
- Revenue counts **SUCCESS** rows only, by Dar es Salaam day.

Windows

- Patterns 1 and 2 and location: **5 minutes**.
- Pattern 3: **7 minutes**.
- Failed withdrawals: **10 minutes**, lock lasts **24 hours**.
- Alerts: FRAUD and score **above 50**; review note **41–70**.

> **In one sentence.** The system blocks money when behaviour looks like a burst, a spray or a mule, when the customer teleports across the country with a large amount, when the hour or the size is extreme, or when the wallet has just failed three withdrawals — and it records a score, a reason and, for every successful payment, the fee that earned the business money.